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ME3330 – Heat Exchangers Technical Summary

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Deliverable 1:

A cross flow heat exchanger runs cold fluid perpendicular to the direction of hot fluid flow. Although not the best for convection, as they lose out to counter flow due to its higher velocity, it remains a more practical option and is the go-to approach for uses such as automotive. Radiators and intercoolers are both forms of heat exchangers and are ubiquitous with engine design. Engineers must design a heat exchanger capable of keeping the engine within ideal operating conditions for maximum efficiency, but without being too costly. This is why knowing the convective heat transfer coefficients for different places in the heat exchanger is crucial for its proper design.

The convective heat transfer coefficient (h) is the rate of heat transfer between a solid and a fluid surface per unit area and temperature. Originating from Newton's law of cooling, h depends on the surface geometry and the fluid's velocity, properties, and the nature of the motion. This final part regarding the nature of the fluid motion will be the focal point of the analysis. The analysis begins with the raw data, and using equation 1, the "ln column" was solved at every point. Equation one is solved using rod temperature $T(t)$, temperature of the air, T_∞, and finally the starting rod temperature T_i which is assumed to be the maximum temperature.

$$\ln\left(\frac{T(t) - T_\infty}{T_i - T_\infty}\right) \tag{1}$$

With all of these points plotted, the convective heat transfer coefficient is approximated by drawing a trendline through the data. This was done with a trendline fitting a linear form of y = mx + b. The results of plotting the "ln column" against time and applying the trendline are below in FIG. 1.

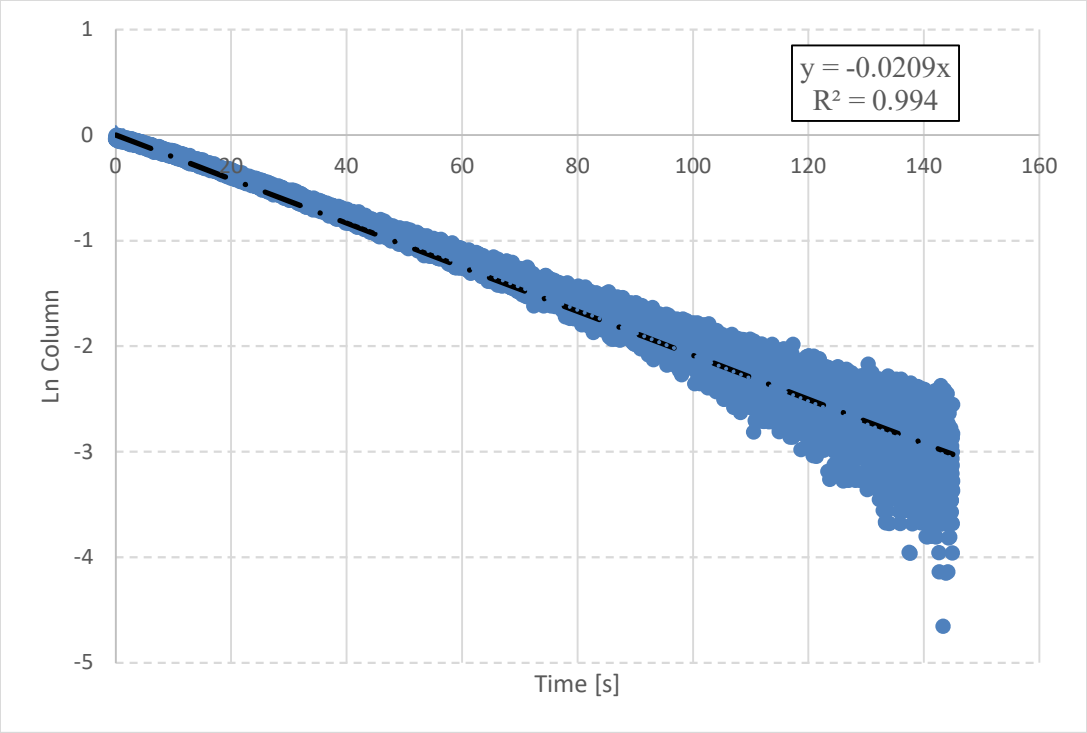


FIGURE 1: Results of equation 1 with trendline.

This calculation was repeated for the top half of the rods, and the results were mirrored to the remaining values. This is because the flow is assumed to be evenly distributed, and the cooling properties uniform. The line of symmetry is visualized in FIG. 2 below. For example, the h value for rod 1 would be experimentally equivalent to the value for rod 5.

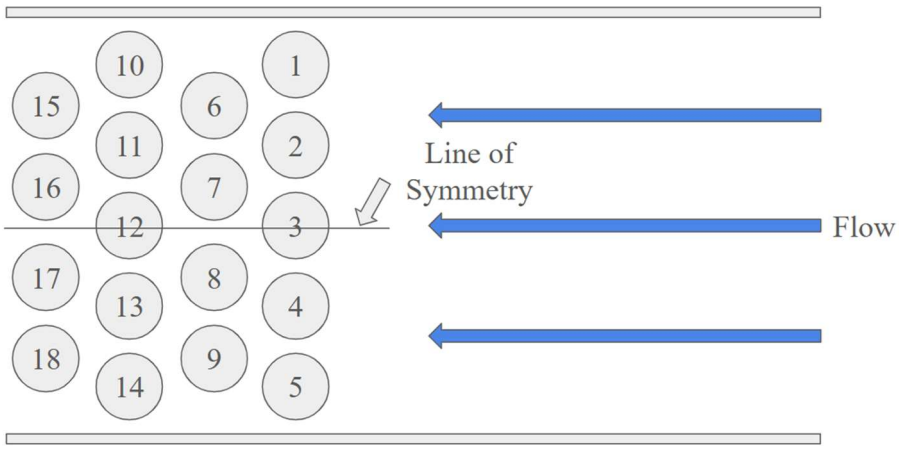


FIGURE 2: Experiment visualization with line of symmetry and flow direction.

The results of applying these calculations to the rods of interest results in the following values, which can be seen in TABLE 1.

Column 4	Column 3	Column 2	Column 1	
265.690064	275.530437	216.4882	165.0996	Row 1
279.9034	272.2503	228.5153	167.2863	Row 2
279.9034	274.9039	228.5153	170.5665	Row 3
265.690064	272.2503	216.4882	167.2863	Row 4
	275.530437		165.0996	Row 5

TABLE 1: Convection heat transfer coefficients for all rods.

The differences in values among a column can be explained by the velocity profile of the air. The air on the walls has lower velocity, and the air in the middle has higher velocity. A higher velocity means more air flowing over the rod, giving it more fluid to spread the heat over. This effect is visualized below in FIG. 3.

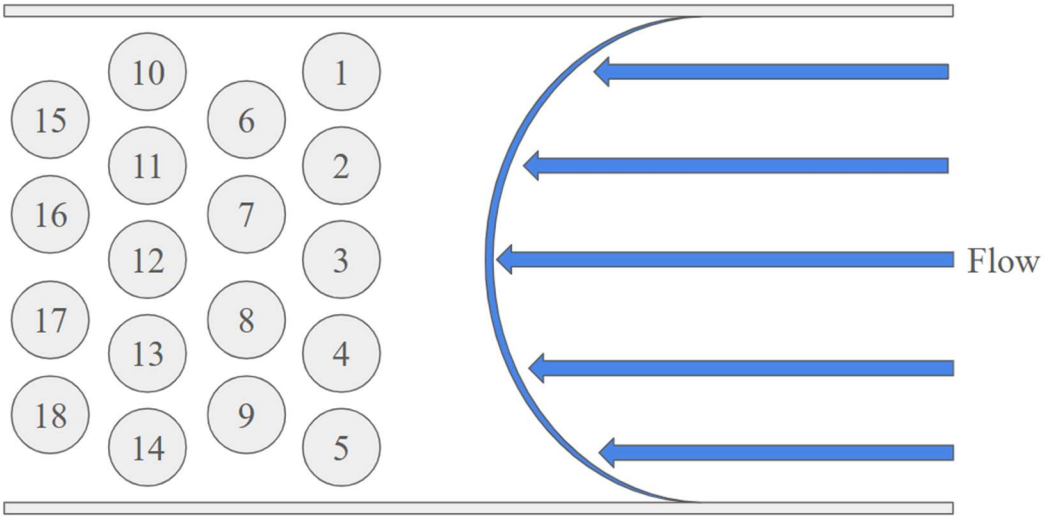


FIGURE 3: Velocity profile of air flowing over rods

The second change in h values occurs throughout a row. Keeping the discussion on velocity, it seems intuitive that the velocity would be highest on the first rods in the airstream, and therefore they have the higher h value. This is contradicted by the experimental results. In general, the h value increases as the air passes through the rods. This is due to the formation of vortexes as the air passes over the first column of rods. This is known as mixing, and with each subsequent occurrence, the h value increases less and less. This is a result of the law of diminishing returns.

To summarize, the highest h values occur in the center of the airflow where velocity is the highest, and at the back of the columns where the mixing has taken full effect. This rod will have the highest convection heat transfer coefficient.

There are inconsistencies within the data, which can be seen in column three. Column three has a lower h value in the middle, which counteracts all other conclusions drawn from the remaining data. This could be a potential outlier, or a result of the strange manner that vortexes behave in.

Deliverable 2:

The velocity profile seen above in FIG. 3 was obtained by first taking pressure measurements from the top to the bottom of the wind tunnel where the rods were placed. These measurements were taken using a pitot tube which records pressure as a displacement of liquid recorded as height. The measurements were taken every 0.005 meters, and the height was used alongside the densities and gravity to solve for the

velocity of the airstream using equation 2.

$$V_\infty = \sqrt{\frac{2(P_s - P_\infty)}{\rho_{air}}} \tag{2}$$

The velocity values were calculated from the top wall to the bottom wall. The resulting values for velocity and distance from bottom wall and are plotted below in FIG. 4

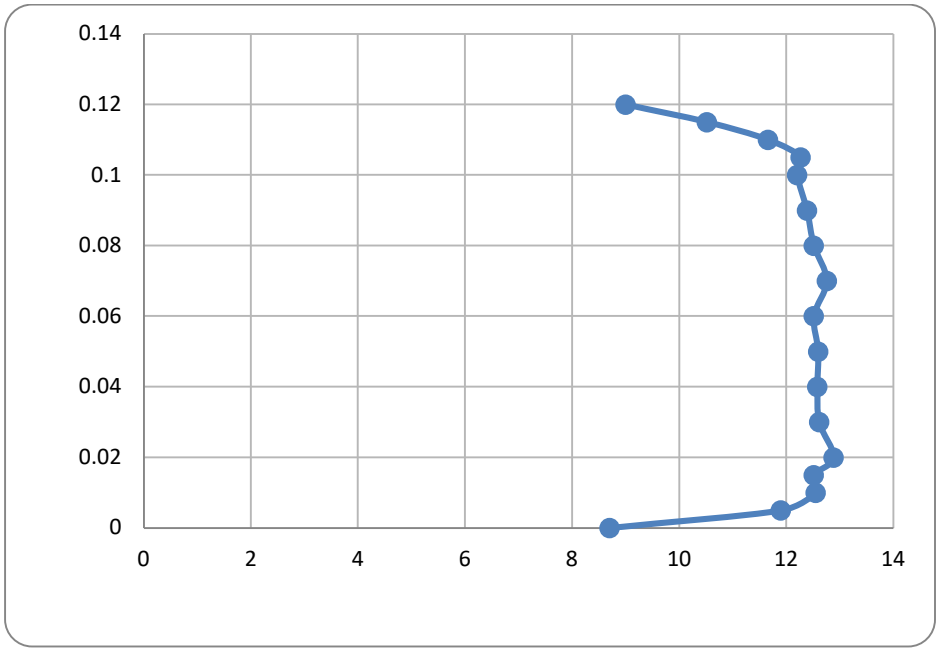


FIGURE 4: Velocity profile of heat exchanger. Velocity on the x [m/s] vs height [m]

It is important to note that the values are flipped relative to all other visualizations used previously. The further right, the higher the value. This goes to show that the velocity profile varies significantly between the wall and the points close to it, before having a reduced impact on the center values. This velocity profile corroborates the previous discussion for the reduction of convection heat transfer coefficient values near the walls. The reduction of velocity can be as large as 4 m/s between the center points and the wall. This helps explain the distribution of experimental values amongst the columns where the highest values are in the center because the boundary layer ensures the slowest velocities are on the walls.

FIG. 4 does not perfectly illustrate the boundary layer due to the inconsistency of the pitot tube. The values jumped around and were not consistent, therefore FIG. 4 should be used to draw conclusions, but does not provide concrete evidence. Based on the experimental values, there should be more distinction among the middle of the distribution around the 0.04-0.08m area. This section featured different values where FIG. 4 suggests that they should be very similar. This is most likely explained by the failure of the pitot tube to accurately record the subtle effect of the boundary layer in this range of values.

Deliverable 3:

Now that there is an understanding of how a heat exchanger works, and its interactions with the boundary layers of airflow, it is time to apply this understanding. A heat exchanger will be designed for a certain heat transfer rate, and the number of columns required to obtain this will be calculated. The heat transfer rate for a given column is found using equation 3, a modified version of Newton's law of cooling.

$$\dot{Q} = n\bar{h}A_s(T_s - T_{in}) \tag{3}$$

This equation works for both types of columns by having the constant n, which represents the number of rows in each column. $\bar{h}$ is the average convection heat transfer coefficient, A_s is the surface are of a rod, T_s is the surface temperature, and T_{in} is the temperature hitting the rod. The average convection heat transfer coefficient is found using equation (4) below.

$$\bar{h} = \sum_{i=1}^{n} \frac{h_i}{n} \tag{4}$$

Due to the aforementioned law of diminishing returns, following the 4th column, $\bar{h}$ will be assumed to be constant due to insignificant changes in its value. A_s will be solved by finding the surface area of the rod exposed to the airstream. This is done below in equation 5.

$$A_s = \pi D_{rod} L \tag{5}$$

Equations 3, 4, and 5 find the rate of heat transfer for one column, but only the T_{in} is known for the first column. Therefore, another equation is required to solve the T_{out} of the column in order to serve as the T_{in} for the next column. This is done using equation 6.

$$T_{out} = T_s + (T_{in} - T_s) * e^{\frac{-n\bar{h}A_s}{\dot{m}c_{p,air}}} \tag{6}$$

The $c_{p,air}$is a constant value measuring the specific heat capacity of the air, and the mass flow rate, $\dot{m}$ is calculated using the velocity and the density of the air. Density is a known constant, but the velocity is solved through approximation by finding the area under the curve of FIG. 4 and using equation 4. With velocity known, mass flow rate can be determined using equation 7.

$$\dot{m} = \rho\overline{\forall} \tag{7}$$

Now it is time to begin solving for the heat transfer rate. The values to be used in this process are included in TABLE 2.

T_{in} [◦C]	T_s [◦C]	$\dot{m}$ [kg/s]	L [m]	$c_{p,air}$ [j/kg*k]	d_{rod} [m]
10	150	0.21594	0.1016	1005	0.0127

TABLE 2: Values for solving equations 3 and 6

The heat exchanger is to be designed for 4000w of heat dissipation into the airstream. The results of the calculations are below in TABLE 3.

Column No.	Rod No.	$\bar{h}$ [W/m^2*K]	T_{in} [◦C]	T_{out} [◦C]	$\dot{Q}$ [w]	$\dot{Q}_{total}$ [w]
1	5	167.06766	10	12.16746092	474.0648239	474.0648239
2	4	222.50175	12.16746092	14.43986147	497.2702534	971.3350773
3	5	274.0930748	14.43986147	17.86594454	753.0908531	1724.42593
4	4	272.796732	17.86594454	20.53182182	584.4686494	2308.89458
5	5	272.796732	20.53182182	23.78866016	715.845841	3024.740421
6	4	272.796732	23.78866016	26.33504339	558.2706978	3583.011119
7	5	272.796732	26.33504339	29.44589871	683.7590991	4266.770218

TABLE 3: Results of heat exchanger calculations.

As seen in TABLE 3, the total heat dissipation reaches the required 4000w in the seventh column. This is the minimum number of rows required to achieve this goal considering all other variables remain the same. What is important to note is that even though the air continued to warm, theoretically making it harder to lose heat, each subsequent row continued to shed more heat up until the final row. The numbers appear to jump up and down, but this is due to the different number of rods found in rows.

Deliverable 4:

To find the uncertainty of the calculations so far, the Kline-McClintock Uncertainty approach will be used. This involves identifying the values that carry uncertainty with them. These are the recorded values of T_s which is T(t), T_i, and T_∞. The uncertainty of T_i, and T_∞ are u_{T_i} and u_{T_∞} respectively. These both have a value of 0.5 ◦C. $u_{T(t)}$ will be found differently from the others. It requires equation 8.

$$u_{T(t)} = \frac{t_{v,p} * S_{yx}}{\sqrt{N_m}} \tag{8}$$

Equation 8 uses $t_{v,p}$, a value from the t-table, the number of measurements N_m, and the standard error S_{yx}. The values used to solve equation 8, as well as the result are in TABLE 4 below.

$t_{v,p}$	S_{yx}	N_m	$u_{T(t)}$ [◦C]
1.96	4.553903	14500	0.074123

TABLE 4: Uncertainty calculations for equation 8

The next step involves taking the equation used to calculate h, and differentiating it with respect to the variables of interest. In this case, the equation will be differentiated with respect to all three variables of interest.

$$\frac{dh}{dT(t)} = \frac{-\rho c_p D_{rod}}{4(T(t) - T_\infty)t} \tag{9}$$

$$\frac{dh}{dT_i} = \frac{\rho c_p D_{rod}}{4(T_i - T_\infty)t} \tag{10}$$

$$\frac{dh}{dT_\infty} = \frac{\rho c_p D_{rod}}{4t}\left(\frac{1}{T(t) - T_\infty} - \frac{1}{T_i - T_\infty}\right) \tag{11}$$

All uncertainty values are listed in TABLE 5.

$\frac{dh}{dT(t)}$	$\frac{dh}{dT_i}$	$\frac{dh}{dT_\infty}$	$u_{T(t)}$ [◦C]	u_{T_i} [◦C]	u_{T_∞} [◦C]
-38.8317	1.466112	37.30282	0.074123	0.5	0.5

TABLE 5: Uncertainty values for solving equation 12

With these equations, they can be combined together using the Kline McClintock method to take the form of equation 12. The equation will be solved at the final point recorded. Before

$$u_h = \sqrt{\left(\frac{dh}{dT(t)}u_t\right)^2 + \left(\frac{dh}{dT_i}u_{Ti}\right)^2 + \left(\frac{dh}{dT_\infty}u_{T_\infty}\right)^2} \tag{12}$$

This equation results in a value of 18.8864 [w/m^2 K], the complete uncertainty of the calculation for h.

One important element of the uncertainty calculation is the inclusion of time in the equation. This means that at the beginning, when a change in time represents a large amount of the total time, errors are large. This is similar to the phenomena in FIG. 1 whereas the rod temperature approaches the air temperature the data becomes more scattered due to small changes having larger impacts.

REFERENCES

[1] Department of Mechanical Engineering, Clemson University. *ME 3330 – Mechanical Engineering Lab II Student Manual.* Fall 2025 ed., Clemson University, 2025.

[2] N Hale, *Heat Exchanger Writing Day* Clemson University 2026,